

Deep Layers

Looking beneath the surface
of a painting

Simone Caglio (707898)
Marco Manduca (804413)



Context & Problem

Problem: Infrared reflectography is a standard, non-invasive conservation tool used to reveal underdrawings, *pentimenti*, and other anomalies beneath the visible layer; however, reading and interpreting the differences between the visible and infrared images requires significant time and specialized expertise on the part of the operators.

Objective: a full pipeline for detection of anomalies in real IR images. A network learns to predict the IR appearance of a painting starting from RGB visible image and wherever the real IR image departs from that prediction, the differences point to something the visible surface does not explain.



Data

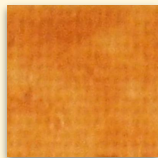
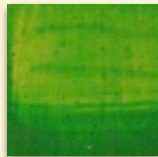
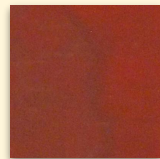
Small dataset: ~1200 images from 25 artworks of different sizes + 6 mockups

Data Augmentation:

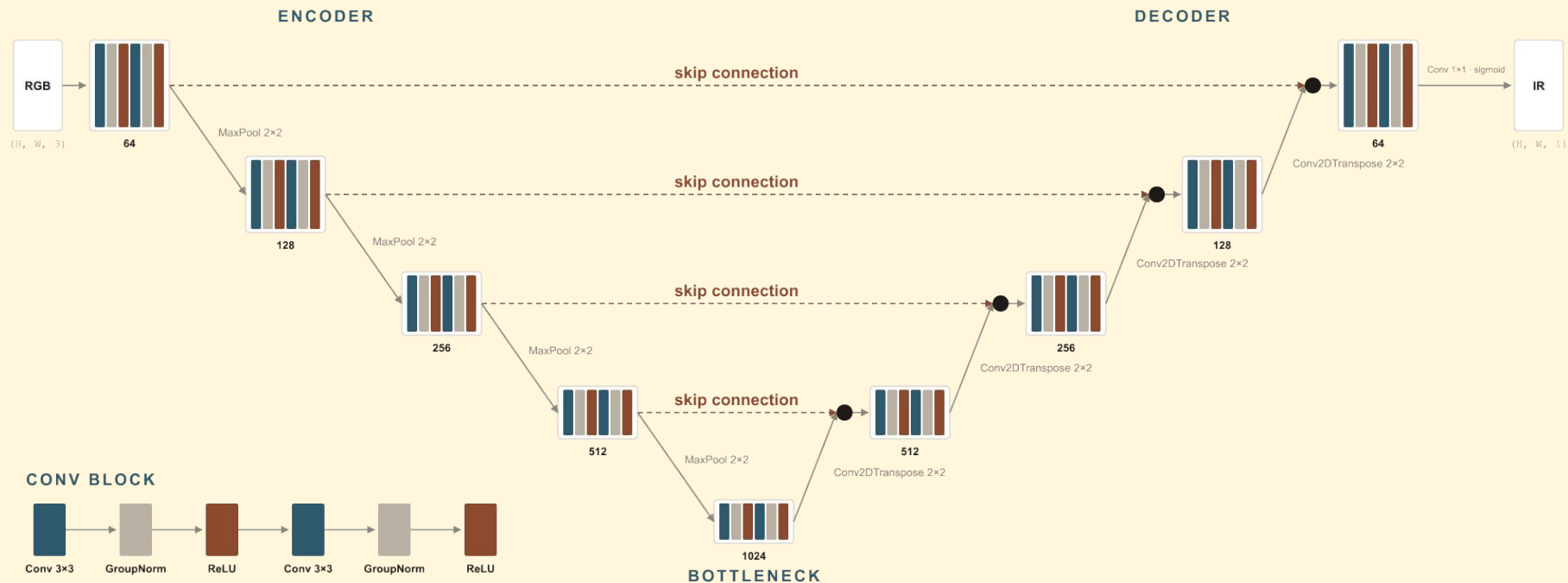
- random horizontal and vertical flip
- same random squared-crop on same pairs of images
- random brightness and contrast modification (RGB only)

Data Splits:

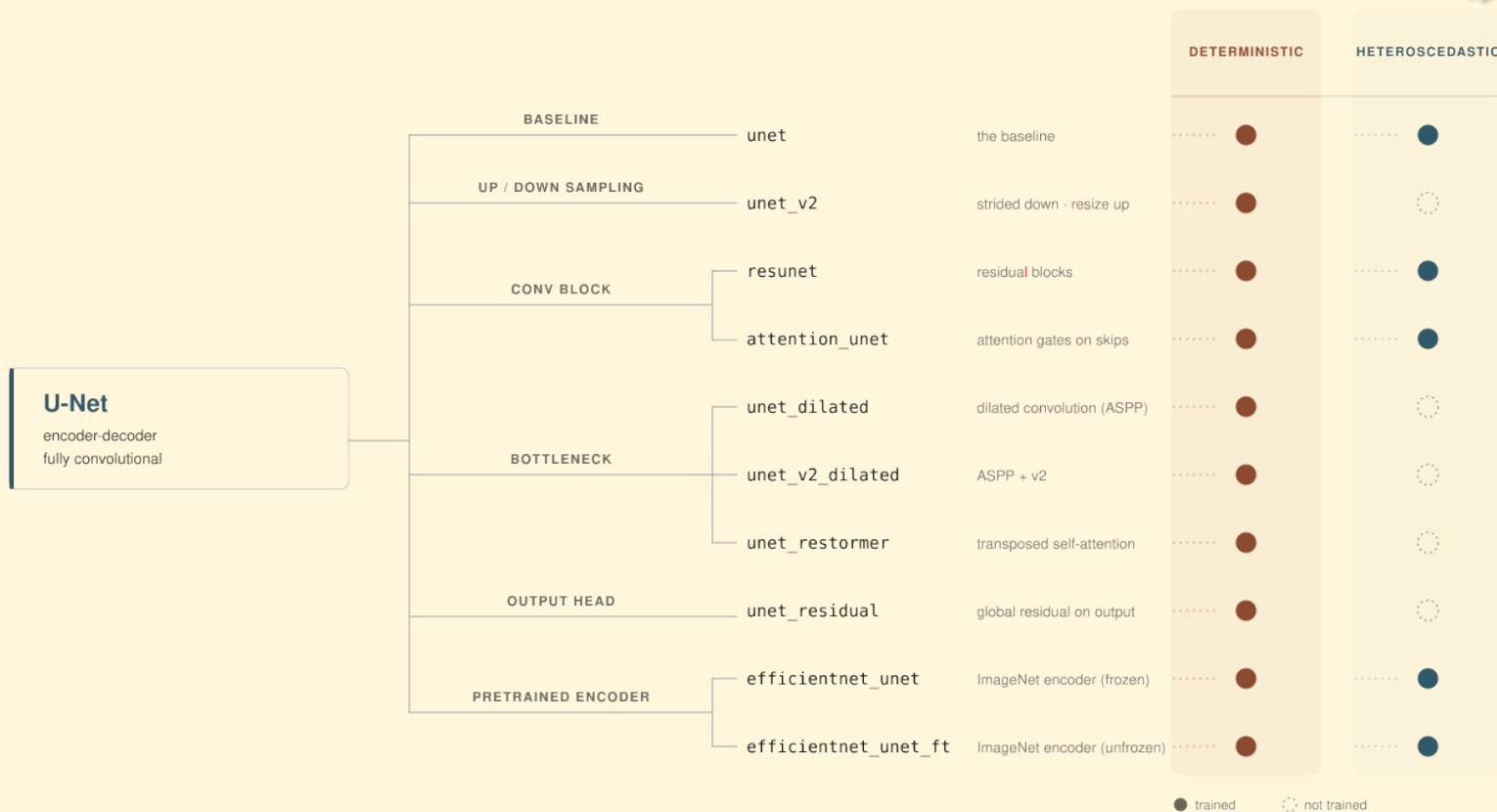
- Holdout (with mockups and same artwork constraints): 70/15/15
- Holdout (no mockups and artworks constraints): 85/15
- Cross Validation: 3 fold



UNET (Model Baseline)



Model Variants and Architectures



Characteristics of the Models

- Architecture: Fully-convolutional encoder-decoder
- Weight initialization: He¹
- Activation Function (hidden Layers): ReLU
- Activation Function (output): Sigmoid
- Optimizer: Adam
- Batch Size: 8
- Normalization: GroupNorm²
- Loss Function (deterministic) : $0.16 \cdot \text{Charbonnier} + 0.84 \cdot (1 - \text{MultiScale-SSIM})$ ^{3,4}
- Loss Function (heteroscedastic): β -weighted Laplace NLL ($\beta=0.5$)⁵

[1] K. He, X. Zhang, S. Ren, J. Sun, "Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification," *ICCV*, 2015.
<https://doi.org/10.48550/arXiv.1502.01852>

[2] Y. Wu, K. He, "Group Normalization," *ECCV*, 2018. <https://doi.org/10.48550/arXiv.1803.08494>

[3] H. Zhao, O. Gallo, I. Frosio, J. Kautz, "Loss Functions for Image Restoration with Neural Networks," *IEEE TCI*, 2017. <https://doi.org/10.48550/arXiv.1511.08861>

[4] W.-S. Lai, J.-B. Huang, N. Ahuja, M.-H. Yang, "Deep Laplacian Pyramid Networks for Fast and Accurate Super-Resolution," *CVPR*, 2017.
<https://arxiv.org/pdf/1704.03915>

[5] M. Seitzer, A. Tavakoli, D. Antic, G. Martius, "On the Pitfalls of Heteroscedastic Uncertainty Estimation with Probabilistic Neural Networks," *ICLR*, 2022.
<https://doi.org/10.48550/arXiv.2203.09168>

Signals for the detection of hidden details

- **Raw delta**, the direct, pixel-by-pixel absolute error:
$$\text{raw_delta} = |\text{real_IR} - \text{pred_IR}|$$
- **Structural delta**¹, where S is the structure component of the SSIM
$$\text{structural_delta} = 1 - S$$
- **Learned z-score**, the error normalized by the uncertainty that the heteroscedastic model assigns to that pixel
$$|z| = |\text{real_IR} - \mu| / \sigma \quad \text{with} \quad \sigma = b \cdot \sqrt{2}$$
- **Structural z-score**, Combines structure and normalization for uncertainty
$$\text{structural_z} = \text{structural_delta} / \text{smoothed}(\sigma)$$

[1] Zhou Wang, A. C. Bovik, H. R. Sheikh and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," IEEE Transactions on Image Processing, 2004, doi: 10.1109/TIP.2003.819861.

Evaluation of signals

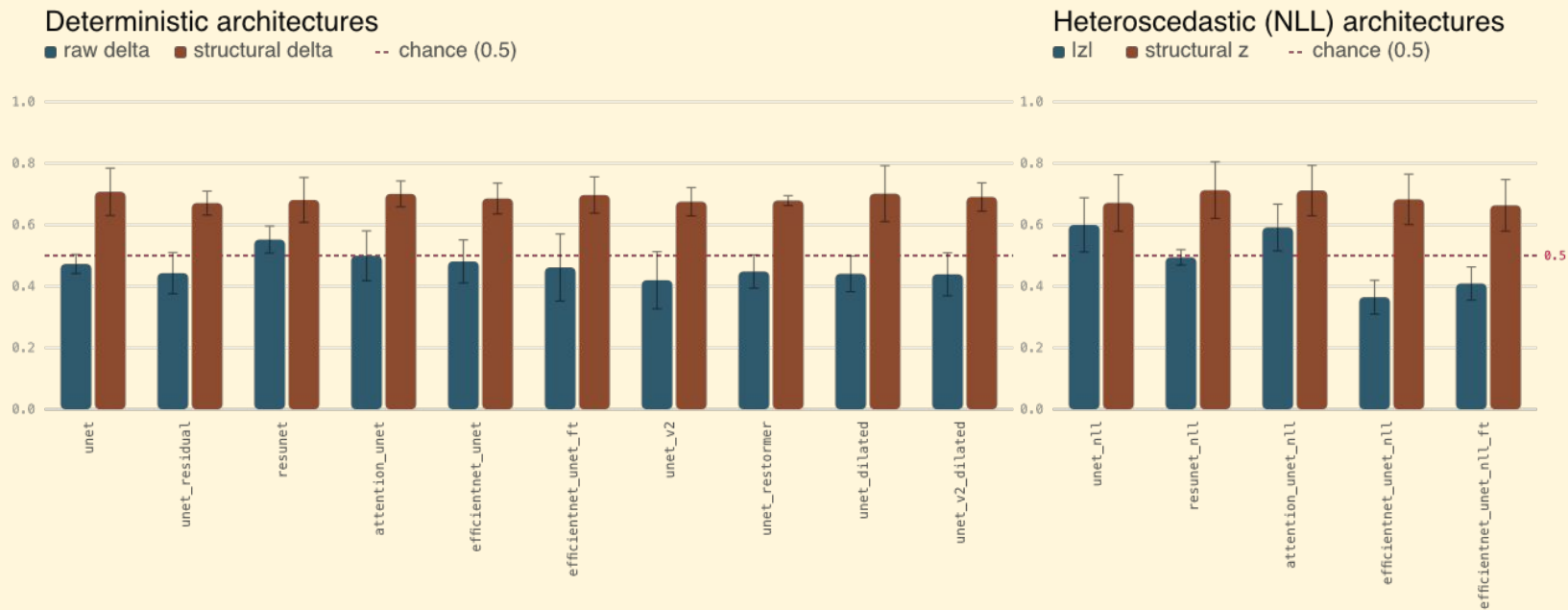
- **AUC** how well the signal ranks positive pixels above negative ones (0.5 = chance, 1.0 = perfect), independent of the percentage of positive pixels in the mask.
- **LIFT**: ratio between Average Precision and prevalence.
- **Stroke coherence**: the extent to which the signal consists of oriented strokes (like a genuine underdrawing) rather than isotropic noise.

Comparison with methods that do not require training:

- linear regression residuals
- band-pass filtering on linear regression residuals
- PCA / ICA

The signals derived from the network are superior.

Results: by metrics (AUC)



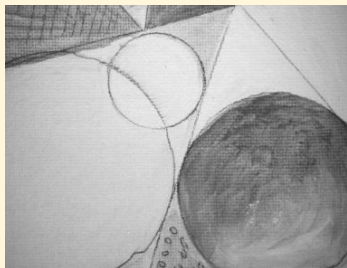
Which is the best?

Results: what worked...

Visible (RGB)



Real IR



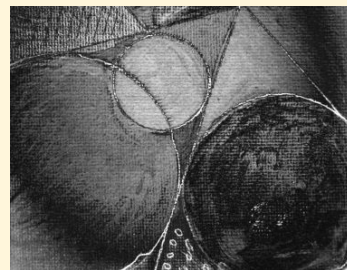
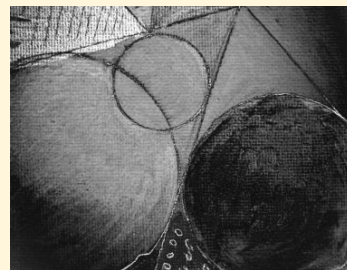
GT Mask



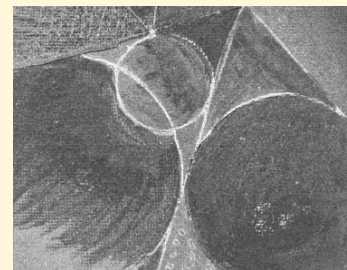
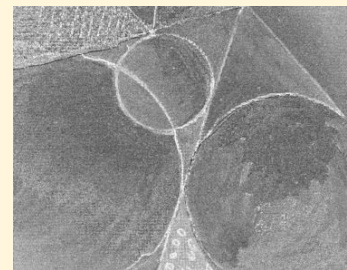
Predicted IR



$|Z|$



Structural Z



resunet

attention_unet

Results: ...and what didn't

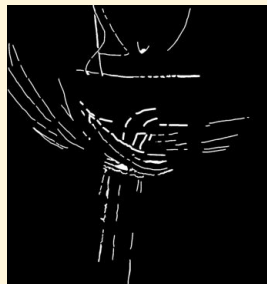
Visible (RGB)



Real IR



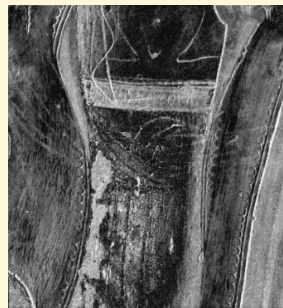
GT Mask



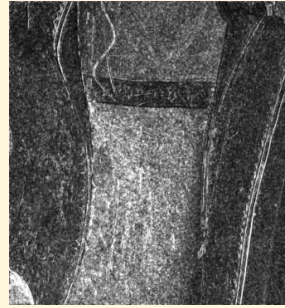
Predicted IR



$|Z|$



Structural Z



resnet

attention_unet

Conclusions

The pipeline produced so far is similar to a student who is still learning:
it **understands** what needs to be done
but still needs to gain a **great deal of experience**...

Open issues:

- increase the dataset (when possible)
- a single IR image contains "flattened" information
(improvement: multi-/hyperspectral images)

Models link

Google drive link to trained models

<https://drive.google.com/drive/folders/1OpMsN6SZ82tpZnXXZyBahYMtctZePkXb?usp=sharing>